

Yogesh Kumar Bandhe

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Research Interests: Digital Forensics, Systems Security, Storage and File System Analysis, Mobile and Embedded System Security, Post-Incident Analysis

Research Profile

Researcher and Software Engineer with strong experience in digital forensics and systems-level security. Currently working as a Junior Research Fellow on forensic disk imaging and NTFS file system analysis, focusing on forensic soundness, integrity verification, and post-incident data recovery. Author of multiple IEEE conference papers on secure and auditable systems, including Best Paper Awards. Research interests center on storage and file-system forensics, system compromise analysis, and security of mobile and embedded systems. Seeking a PhD position in systems security and digital forensics.

Education

B.Tech. in Computer Science Engineering, Bhilai Institute of Technology, Durg
2019–2023 — CGPA: 8.99/10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems

Current Research Experience

Forensic Disk Imaging and NTFS File System Analysis National Institute of Technology Raipur

Junior Research Fellow (JRF) 2025–Present

- Designing and implementing a forensic-grade disk imaging and analysis tool for Windows-based storage devices.
- Implementing bit-stream acquisition of physical and logical drives with integrity verification using MD5 and SHA-1/SHA-256 hashing.
- Supporting multiple forensic image formats including RAW (DD), E01, and AFF.
- Parsing NTFS structures such as Boot Sector, \$MFT, \$Bitmap, runlists, and timestamp attributes for timeline reconstruction and file recovery.
- Recovering deleted artifacts and reconstructing file metadata for post-incident forensic analysis.
- Integrating chain-of-custody metadata logging to ensure forensic soundness, repeatability, and evidentiary integrity.
- Developing a controlled acquisition and analysis interface using Python and PyQt, emphasizing validation and forensic best practices.

Research Projects

Secure and Auditable IoT Payment System (Shramik Bhugtaan Seva Kendra – SBSK) IIT Bhilai Innovation and Technology Foundation

Project Engineer 2023–2025

- Designed a secure IoT-enabled framework for transparent wage payment and work monitoring in rural welfare schemes.
- Implemented two-factor authentication using NFC smart cards and real-time facial recognition (Python, OpenCV).
- Developed long-range LoRaWAN communication with ACK-based chunked transmission for low-connectivity environments.

- Built RESTful APIs using Flask for secure cloud-based transaction processing, logging, and audit trails.
- **Paper presented** at ICSCA-2025; received the **Best Paper Award**.

Solar-Powered IoT Payment Solution

IIT Bhilai

IoT Research Project

2023–2025

- Developed solar-powered IoT devices for reliable and fault-tolerant digital payments in off-grid regions.
- Integrated MongoDB and Firebase for real-time transaction processing and synchronized record keeping.
- Emphasized low-power design, system reliability, and resilience in constrained environments.

Application Development Experience

Pokemon Explorer App

Android (Kotlin, Jetpack Compose)

Offline-capable data explorer using PokeAPI and modern Android architecture components.

Secure Note Application

Android, Firebase

Secure, real-time synchronized note-taking application supporting 500+ users.

Thermal Printer Desktop Application

JavaFX

Desktop application supporting ESC/POS commands for receipt generation and thermal printers.

Technical Skills

- **Digital Forensics & Systems Security:** Disk Imaging, NTFS Internals, File System Analysis, Hashing, Chain of Custody
- **Programming Languages:** Python, Java, Kotlin, C, SQL
- **Operating Systems & Storage:** Windows Internals, File Systems, Low-Level Data Structures
- **IoT & Embedded Systems:** ESP8266, LoRa/LoRaWAN, Sensors, Secure Communication
- **Frameworks:** Flask, FastAPI, Android SDK, Jetpack Compose, JavaFX
- **Databases:** MongoDB, Firebase, MySQL, SQLite
- **Tools:** Git, Postman, Android Studio

Publications

1. Timely Payment to the Beneficiaries Residing in Far-Flung Areas using IoT and Cloud Database Technology.
IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI 2025).
2. Solar Powered IoT Solution for Timely Payments to Rural Beneficiaries Under Government Schemes.
IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI 2025).
Presented by Author; Best Paper Award.
3. An IoT-Enabled Framework for Transparent Wage Payment and Work Monitoring in Rural Welfare Schemes.
Second International Conference on Systems, Control and Automation (ICSCA-2025).
Presented by Author; Best Paper Award.

Achievements

- Best Paper Award, ICSCA-2025 (International Conference)
- Qualified GATE 2025 (Computer Science & IT)
- Top 0.1% in PrepSAT Hackathon (Rank 65 / 90,678)
- CG PET Rank: 252

Certifications

- Java & SQL — PrepInsta
- Python — Coursera
- Android Development — Internshala